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Econometrics

How does the belief that success can be achieved through hard work rather than luck affect life satisfaction in a country?

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Introductory part describing the formulation and rationale of the question

Research area

Social and humanitarian sciences

Formulation research question

How does the belief that success can be achieved through hard work rather than luck affect life satisfaction in a country?

This study aims to explore the relationship between cultural values and subjective well-being at the macro level. We test the hypothesis that the collective belief in "hard work" as the key to success acts as an important social resource, the value of which varies depending on the objective conditions in the country.

Our analysis focuses on the question of whether this value is becoming a more significant factor in life satisfaction in societies that face more difficult conditions that can affect satisfaction and faith in success (for example, economic instability, weak institutions and high levels of inequality). In other words, we are exploring whether a shared belief in the power of work can serve as a stabilizing factor, helping entire nations maintain a higher level of well-being despite systemic difficulties.

At the level of aggregated data, we explore the main relationship by monitoring key macroeconomic and social indicators of countries. The model includes GDP per capita, the Gini index, the unemployment rate, and the share of health care costs in GDP. This makes it possible to isolate the effect of faith in hard work from the influence of the country's total wealth, economic inequality, the state of the labor market and the quality of the healthcare system, thereby providing a more reliable test of the hypothesis about its role as a factor of collective sustainability.

Thus, we empirically test whether the link between faith in work and life satisfaction is universal for all countries, or whether its strength depends on the macro context: in unfavorable countries, this link is strengthened, whereas in prosperous societies, where success may be perceived as something more guaranteed, it may be weaker.

Actuality

In the context of global crises, growing inequality and economic instability, the question of which cultural resources help entire societies maintain stability and well-being is becoming particularly acute. There are widespread beliefs that *"success comes through hard work", "you can't even catch a fish out of a pond without effort", "patience and hard work will overcome everything"*, which face macroeconomic realities where results are determined not only by collective diligence, but also by structural factors: the state of the economy, the level of inequality and the effectiveness of social institutions.

Our research is relevant because it shifts this issue from the plane of individual psychology to the sphere of national policy and social management. By empirically testing how faith in the power of work is related to the level of happiness in countries with varying degrees of disadvantage, we can determine whether this value is a universal asset or whether its role depends on the context. These findings are important for understanding which cultural mechanisms can be used in shaping public policy aimed at strengthening social cohesion and improving the quality of life in difficult economic conditions.

Description of data

The data for the research were taken from two sources. The first source is the World Values Survey (WVS) [1] – a large-scale cross-national survey that was provided in the list of allowed data sources. The second source is World Bank Data (WBD), specifically the World Development Indicators (WDI) database [2]. For the analysis our team combined WVS Waves 5, 6, 7: Wave 5 (around 2005–2009), Wave 6 (around 2010–2014), Wave 7 (2017–2022) with macroeconomic indicators from the WDI. The unit of analysis in the final dataset is the country–wave: for each country and each wave we calculate average values of the key attitudinal variables from the WVS and merge them with country-level economic indicators from the WDI.

Table 1: Description of Variables and Data Sources

	First Name	Last Name	Label
	WVS		
	B_COUNTRY, V2	COUNTRY	Country name (after conversion from numeric WVS codes)
Temporal control variable	V260, V262, A_YEAR	Wave 5, Wave 6, Wave 7	Around 2005–2009, 2010–2014, 2017–2022
Dependent variable	V22, V23, Q49	satisfaction5, satisfaction6, satisfaction7	Life Satisfaction for Waves 5, 6, 7
Independent variable	V120, V100, Q110	hardwork5, hardwork6, hardwork7	Belief in hard work (1 - the greatest belief in hard work)
Control variables	V46, V55, Q48	control5, control6, control7	Average perceived freedom of choice and control over one's life
	WDI		
	Education5, Education6, Education7		Compulsory education, duration (years)
	industry5, industry6, industry7		Employment in industry (% of total employment)
	GDP5, GDP6, GDP7		GDP per capita
	Gini5, Gini6, Gini7		Income inequality: Gini index
	Unemployment5, Unemployment6, Unemployment7		Unemployment indicator
	Health5, Health6, Health7		Health expenditure indicator

V22, V23, Q49 variable - life satisfaction for Waves 5, 6, 7 respectively. The satisfaction5, satisfaction6 and satisfaction7 variables measure average life satisfaction in each country for WVS Waves 5, 6 and 7 respectively. This is a continuous variable that measures the respondent's overall subjective well-being. It is based on the question: *"All things considered, how satisfied are you with your life as a whole these days?"* Measurement scale ranges from 1 ("Completely dissatisfied") to

10 ("Completely satisfied). Then, for each country and each wave we compute the mean value of life satisfaction across all respondents. Thus, satisfaction5, satisfaction6, satisfaction7 – average life satisfaction in Wave 5 (around 2005–2009); Wave 6 (around 2010–2014); Wave 7 (2017–2022) respectively.

Q110 variable - belief in hard work. The hardwork5, hardwork6 and hardwork7 variables capture average beliefs about whether success in life depends on hard work or on luck and connections, in each country and wave. At the individual level, the underlying item contrasts the statement that hard work usually brings a better life in the long run with the view that hard work does not generally bring success and that success depends more on luck and connections. Respondents evaluate their position on a 1–10 scale, where lower values correspond to stronger belief in hard work as the main driver of success, and higher values reflect stronger belief in luck and connections. In the code, the following wave-specific variables are used:

Wave 5: V120

Wave 6: V100

Wave 7: Q110

For each wave, negative codes are removed and the country mean of this scale is calculated:

hardwork5, hardwork6, hardwork7 – average belief that success depends on hard work rather than luck in Wave 5; Wave 6; Wave 7 respectively.

Q56 variable - perceived freedom of choice and control. The control5, control6 and control7 variables capture average perceived freedom of choice and control over one's life in each country and wave. The underlying question (with stable wording across waves) is: *"Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them".* Respondents answer on a 1–10 scale, where 1 means no choice at all and 10 means a great deal of choice. In the code, the following wave-specific variables are used:

Wave 5: V46

Wave 6: V55

Wave 7: Q48

After removing non-substantive codes, the mean value is computed for each country and wave: control5, control6, control7 – average perceived freedom of choice and control in Wave 5; Wave 6; Wave 7 respectively.

Macroeconomic control variables from WBD/WDI:

The Education5, Education6, Education7 variables are based on the WDI column *Compulsory education, duration (years)* and indicate the total number of years of compulsory schooling at the country–wave level. For each country:

Education5– mean of the education indicator over 2005–2009;

Education6– mean over 2010–2014;

Education7– mean over 2017–2022

The industry5, industry6, industry7 variables are based on the WDI column *Employment in industry (% of total employment)* and indicate the share of a country's total employed population that works in the industrial sector. For each country:

industry5 – mean of the employment indicator over 2005–2009;

industry6 – mean over 2010–2014;

industry7 – mean over 2017–2022

The GDP5, GDP6 and GDP7 variables represent country-level GDP per capita indicators derived from the World Development Indicators (WDI). In the raw WDI file these values are stored in the column *GDP_per_capita* (current US$). For each country and for each period corresponding to a WVS wave, annual values are averaged as follows:

GDP5 – mean GDP per capita over 2005–2009 (Wave 5);

GDP6 – mean GDP per capita over 2010–2014 (Wave 6);

GDP7 – mean GDP per capita over 2017–2022 (Wave 7).

The Gini5, Gini6 and Gini7 variables measure income inequality using the Gini index from the WDI. In the raw file they appear in the column *Gini_index*. For each country and wave, the annual Gini values are averaged over the same periods as above:

Gini5 – mean Gini index over 2005–2009;

Gini6 – mean Gini index over 2010–2014;

Gini7 – mean Gini index over 2017–2022.

The Gini index typically ranges from 0 (perfect equality) to 100 (maximal inequality).

The Unemployment5, Unemployment6 and Unemployment7 variables are constructed from the WDI column *Unemployment* (% of total labor force) and capture an unemployment indicator at the country–wave level. For each country:

Unemployment5 – mean of the unemployment indicator over 2005–2009;

Unemployment6 – mean over 2010–2014;

Unemployment7 – mean over 2017–2022.

The Health5, Health6 and Health7 variables are based on the WDI column *Health_expenditure* (% of GDP) and represent a health spending indicator at the country–wave level. For each country:

Health5 – mean of the health expenditure indicator over 2005–2009;

Health6 – mean over 2010–2014;

Health7 – mean over 2017–2022.

Data collection and processing

The empirical base of the study is based on two sources: microdata from the World Values Survey (WVS) (waves 5, 6 and 7) and macroeconomic indicators from the World Bank Data (World Development Indicators, WBD/WDI).

At the first stage, we selected the relevant variables from the WVS dataset (life satisfaction, belief in hard work, perceived control over one's life, year of interview, and country). Erroneous and invalid values (including technical codes and negative values) were removed from the survey data. Next, for each wave, individual responses were aggregated to the country–wave level by computing country means. The resulting tables for Waves 5, 6, and 7 were then merged into a single panel dataset.

To construct the balanced panel, the following approach to missing values was applied. If, for a given country and a given indicator, a missing value occurred in only one of the three waves, this value was replaced by the average of the corresponding indicator in the other two waves for that country. In cases where missing values occurred in two or more waves, the corresponding observations were excluded. In this way, we obtained a balanced sample of countries that are observed in all three waves without missing data on the key variables.

In parallel, an unbalanced panel was constructed in which missing values for the same indicators were not imputed but retained as missing. Restrictions were imposed only on completely absent observations (when a variable was missing in all three waves), which were excluded from the analysis. This procedure made it possible to increase the number of countries in the sample by allowing incomplete time information.

At the second stage, we used World Bank Data (WBD/WDI) on GDP per capita, the Gini index, unemployment, public health expenditure, education and employment in the industrial sector. Obvious errors and invalid values were also removed from these data, after which the indicators were averaged over years within intervals corresponding to the temporal coverage of the WVS waves (for Waves 5, 6, and 7). The macroeconomic indicators were then merged with the previously constructed WVS panels by country.

As a result, we obtained two final datasets: a balanced dataset (without missing values on key variables) and an unbalanced dataset (with partial missingness). Since the initial number of observations was limited, the analysis used data from all three waves, and results based on the balanced and unbalanced samples were compared, which allows us to assess the robustness of the findings to different strategies of handling missing data.

Intelligence analysis of data

Sample representativeness

The sample consists of 50 countries and captures broad global variation in economic development, geographic location, and institutional context. The number of time intervals equals 3 waves and the total number of observations is 150.

By level of development, the sample includes:

1. Advanced economies such as the United States, Germany, Japan, and Sweden
2. Developing countries including Brazil, India, China, and Nigeria
3. Transition economies, for example Russia, Ukraine, and Kazakhstan

By geographic coverage, the sample includes multiple world regions:

1. Europe (18 countries)
2. Asia (14 countries)
3. Latin America (7 countries)
4. Africa (7 countries)
5. North America (2 countries)
6. Australia and Oceania (2 countries)

Such geographic diversity allows the analysis to incorporate cross-regional differences in culture, economic development, and institutional quality.

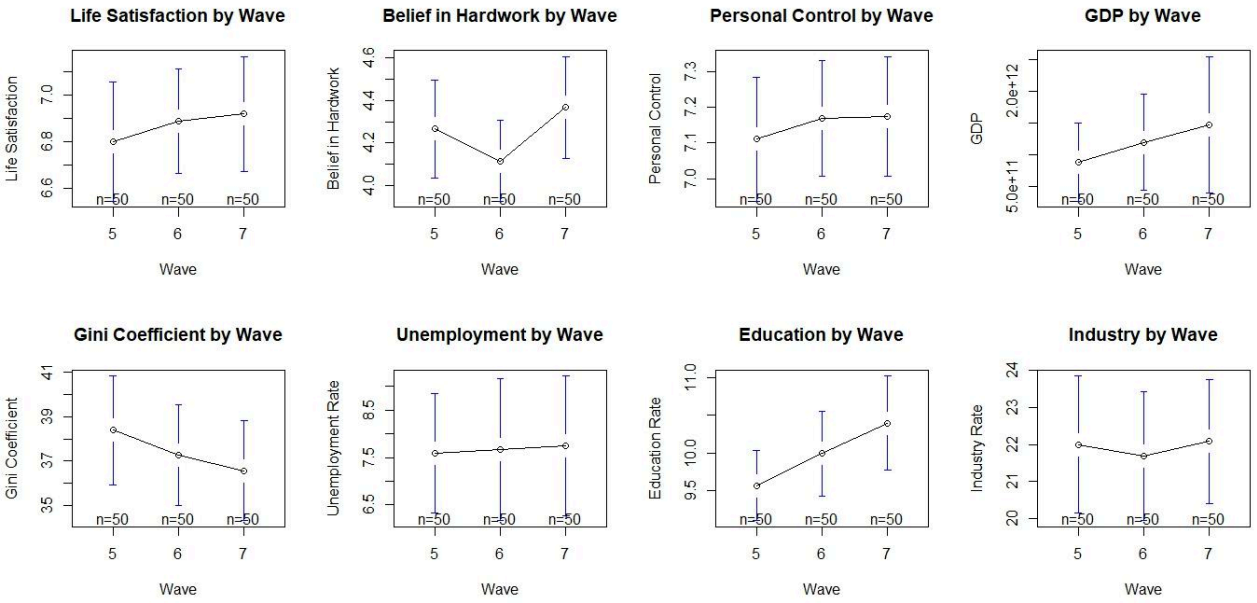
By cultural and institutional context, the sample includes:

1. Both Western and non-Western societies
2. Democracies as well as authoritarian regimes
3. Countries with varying degrees of income inequality and economic stability

This diversity is important, as beliefs about success, individual effort, and the role of luck may be influenced by cultural values, political systems, and socioeconomic conditions.

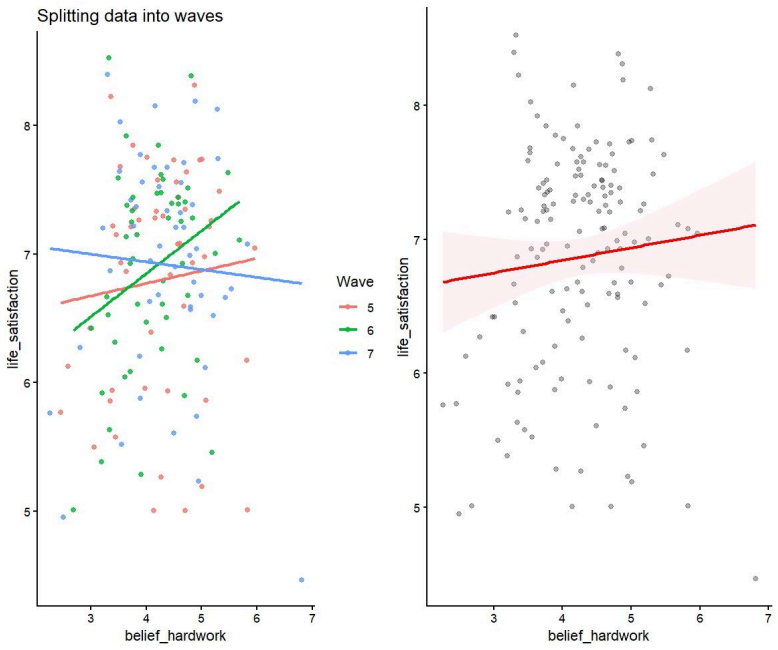
Visualization and expected results

Figure 1: The mean values of variables across the waves



For preliminary data analysis, let's examine the average values of the variables across the waves. A trend can be observed: over time, average life satisfaction increases, as do personal control, GDP and education. Meanwhile, the unemployment rate remains almost unchanged, while belief in hardwork and employment in industry drops in 2010–2014 (wave 6).

Figure 2: Belief in Hard Work by Wave



When analyzing the scatterplot graphs, it is difficult to predict the relationship between belief in hard work and life satisfaction. We observed a positive relationship in 2005–2009 and in 2010–2014, but a negative relationship in wave 7 (2017–2022). It can be assumed that the relationship will be weak or nonlinear.

Formulation of the models and Regression analysis results

Balanced data[1]

Analysis Strategy Description

Models 1-1.3[2]

The analysis begins with a baseline model containing macroeconomic indicators (GDP, inequality (index Gini), unemployment, health expenditure). Subsequently, the following variables are added sequentially:

- Duration of compulsory education

- Share of employment in industry

- Perceived personal control

This approach tests whether the effect of belief in hard work persists after accounting for structural and psychological factors.

For each model specification, three variants are estimated:

- Pooled OLS (without fixed effects)
- Model with country fixed effects (accounts for time-invariant national characteristics)
- Model with country and wave fixed effects (additionally controls for common temporal trends)

Regression Equations

Model 1 (Baseline):

$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot gdp + \beta_3 \cdot gini + \beta_4 \cdot unemployment + \beta_5 \cdot health + \varepsilon$

Model 1.1 (with education): Adds $+ \beta_2 \cdot education$

Model 1.2 (with education and industry): Adds $+ \beta_2 \cdot education + \beta_3 \cdot industry$

Model 1.3 (with education, industry and personal control): Adds $+ \beta_2 \cdot personal_control + \beta_3 \cdot education + \beta_4 \cdot industry$

Models with fixed effects include the components $\alpha_country$ and γ_wave.

Hypotheses Tested

For all models, the following is tested:

H_0: $\beta_1 = 0$ (belief in hard work does not affect life satisfaction)

[1] The models and results for unbalanced data are similar, and to shorten the report, we have added them to the applications

[2] The full version of the models with the introduction of control variables step by step and the introduction of fixed effects step by step with code and technical results can be viewed in the appendix

H_1: $\beta_1 \neq 0$ (a statistically significant effect exists)

Model 1.3 examines whether the effect of belief in hard work is direct or mediated through individual perceived control.

MODEL 1

Regression Equation for Model 1

$life_satisfaction = \beta_0 + \beta_1(belief_hardwork) + \beta_2(gdp) + \beta_3(gini) + \beta_4(unemployment) + \beta_5(health) + \alpha_country + \gamma_wave + \varepsilon$

Coefficient Interpretation:

β_1 - Change in life satisfaction associated with a one-unit increase in belief in hard work (negative values indicate inverse relationship)

β_2 - Effect of GDP per capita on life satisfaction

β_3 - Effect of income inequality (Gini coefficient) on life satisfaction

β_4 - Change in life satisfaction associated with a one-percentage-point increase in unemployment rate

β_5 - Effect of health expenditure on life satisfaction

$\alpha_country$ - Country fixed effects (captures time-invariant country characteristics)

γ_wave - Wave fixed effects (captures common time trends across all countries)

H_0: $\beta_1 = 0$ (Belief in hard work has no effect on life satisfaction after controlling for personal control)
H_1: $\beta_1 \neq 0$ (Belief in hard work affects life satisfaction after controlling for personal control)

Table 2: Estimated Regression Models for Life Satisfaction for Model 1

	Model 1	**Model 1|Country**	**Model 1|Country + wawe**
(Intercept)	5.327***		
	(0.828)		
belief_hardwork	0.049	-0.198+	-0.208*
	(0.146)	(0.099)	(0.099)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
gini	0.031*	-0.008	0.007
	(0.014)	(0.018)	(0.022)
unemployment	-0.074***	-0.038*	-0.043**
	(0.020)	(0.016)	(0.016)

health	0.113**	0.004	-0.028
	(0.037)	(0.048)	(0.066)
Num.Obs.	150	150	150
R2 Adj.	0.225	0.844	0.847
R2 Within		0.167	0.176
R2 Within Adj.		0.123	0.131
RMSE	0.73	0.27	0.26
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

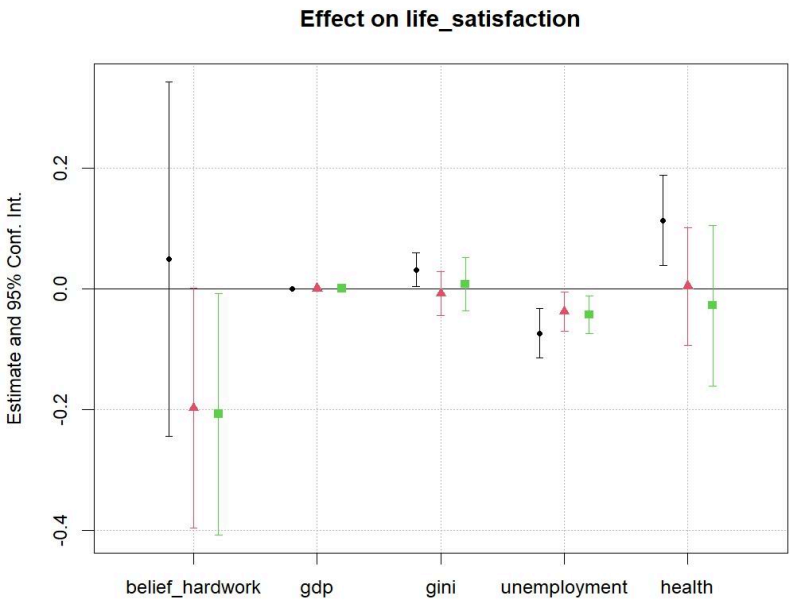
Data from the table:

Pooled OLS (Model 1): belief_hardwork = 0.049 (not significant, p > 0.1)

Country FE (Model 1|Country): belief_hardwork = -0.198+ (p < 0.1)

Country+Wave FE (Model 1|Country + wave): belief_hardwork = -0.208* (p < 0.05)

Figure 3: Confidence Intervals for Model 1



Interpretation:

Without accounting for country differences (Pooled OLS): There is no statistically significant relationship between the average belief in hard work in a country and the average level of life satisfaction. The coefficient is close to zero and insignificant.

When controlling for time-invariant country characteristics (Country FE): When analyzing within-country changes, a negative relationship emerges at the margin of significance (p <

0.1). This suggests that when belief in hard work strengthens in the same country (the variable's value decreases, since 1 = "hard work", 10 = "luck/connections"), life satisfaction tends to increase.

With additional control for time trends (Country+Wave FE): The effect strengthens and becomes statistically significant at the standard 5% level ($p < 0.05$). The coefficient of -0.208 indicates that a one-point increase in the average belief in "luck/connections" (corresponding to a weaker belief in hard work) is associated with a decrease of 0.208 points in the country's average life satisfaction, all else being equal.

Hypothesis for belief_hardwork:

H_0: $\beta_1 = 0$ — rejected in models with fixed effects.

The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Conclusion: The significant relationship only emerges when analyzing within-country dynamics. Cross-country comparisons without fixed effects mask this dependency.

Model 1.1

Purpose of the model: To test the robustness of the effect of belief in hard work after adding control for the level of formal education (education). The model examines whether the initial relationship is confounded by differences in human capital across countries.

Added variable: education (duration of compulsory education in years).

Model specification:

$$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot education + \beta_3 \cdot gdp + \beta_4 \cdot gini + \beta_5 \cdot unemployment + \beta_6 \cdot health + \alpha_country + \gamma_wave + \varepsilon$$

Table 3: Estimated Regression Models for Life Satisfaction for Model 1.1

	Model 1 with education	**Model 1 with education|FE1**	**Model 1 with education|FE2**
(Intercept)	4.049***		
	(0.738)		
belief_hardwork	0.051	-0.202*	-0.208*
	(0.123)	(0.098)	(0.100)
education	0.177***	0.026	-0.001
	(0.049)	(0.047)	(0.047)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
gini	0.025+	-0.004	0.007
	(0.013)	(0.023)	(0.025)
unemployment	-0.057**	-0.039*	-0.043**
	(0.020)	(0.016)	(0.016)

health	0.050	0.000	-0.028
	(0.039)	(0.051)	(0.066)
Num.Obs.	150	150	150
R2 Adj.	0.360	0.843	0.846
R2 Within		0.170	0.176
R2 Within Adj.		0.117	0.122
RMSE	0.66	0.26	0.26
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

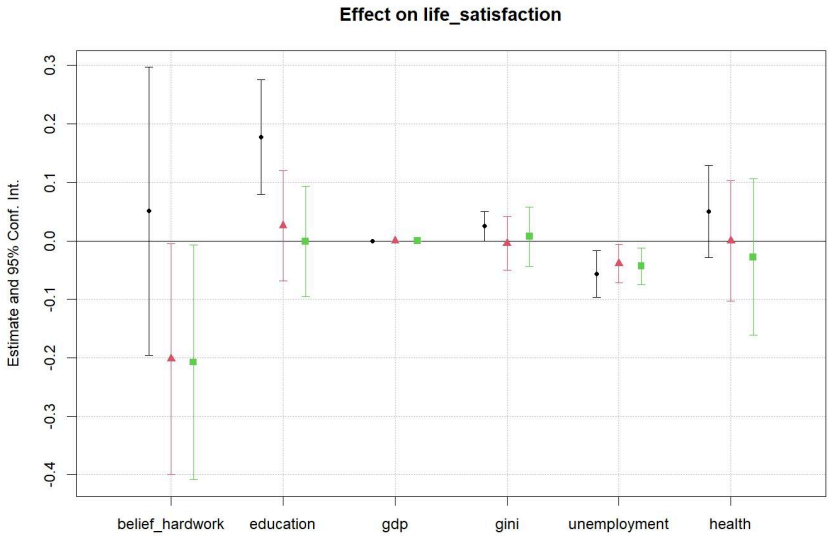
Results and interpretation:

Pooled OLS: belief_hardwork = 0.051 (not significant, p > 0.1)

Country FE: belief_hardwork = -0.202* (p < 0.05)

Country+Wave FE: belief_hardwork = -0.208* (p < 0.05)

Figure 4: Confidence Intervals for Model 1.1



Interpretation:

Effect of Belief in Hard Work (belief_hardwork): The results are almost identical to Model 1. The coefficient remains statistically insignificant in the pooled OLS specification but becomes significant (p < 0.05) and negative in both fixed effects models. The addition of the education control does not change the size or significance of the main effect.

Effect of Education (education): In the pooled OLS model, education has a strong positive effect (0.177***). However, this effect completely disappears in the fixed effects models (coefficients become 0.026 and -0.001, both insignificant). This indicates that the observed

relationship between education and life satisfaction at the country level is driven by stable differences between countries rather than changes within countries over time.

Test of Hypothesis for belief_hardwork:

H_0: $\beta_1 = 0$ — Rejected in both fixed effects models ($p < 0.05$).

H_1: $\beta_1 \neq 0$ — Supported. A statistically significant negative relationship exists between the collective belief in luck/connections (vs. hard work) and national life satisfaction, even after controlling for the duration of compulsory education.

The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Conclusion: The negative effect of a declining belief in hard work (or increasing belief in luck) on national life satisfaction is robust to the inclusion of an education control. The effect is not explained by differences or changes in the level of formal education across countries. The main driver of life satisfaction changes within countries remains the shift in collective beliefs about the determinants of success.

Description of Model 1.2 (Model 1.1 + Economic Structure)

Purpose of the model: To test whether the relationship between belief in hard work and well-being is spurious due to differences in the economic structure of countries. The model tests the robustness of the main effect.

Added variable: industry — the share of employment in the industrial sector (% of total employment). This variable allows for isolating the effect of the cultural value from the effect of the level of industrialization and type of economy.

Model specification:

$$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot education + \beta_3 \cdot industry + \beta_4 \cdot gdp + \beta_5 \cdot gini + \beta_6 \cdot unemployment + \beta_7 \cdot health + \varepsilon$$

Table 4: Estimated Regression Models for Life Satisfaction for Model 1.2

	Model 1 with education and industry	**Model 1 with education and industry|FE1**	**Model 1 with education and industry|FE2**
(Intercept)	3.144***		
	(0.722)		
belief_hardwork	0.007	-0.225*	-0.225*
	(0.118)	(0.094)	(0.096)
education	0.164***	0.021	-0.003
	(0.036)	(0.045)	(0.045)
industry	0.040**	0.036*	0.033*
	(0.013)	(0.015)	(0.014)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)

gini	0.033**	0.004	0.013
	(0.011)	(0.021)	(0.024)
unemployment	-0.068***	-0.035*	-0.040*
	(0.017)	(0.017)	(0.017)
health	0.077*	0.040	0.011
	(0.038)	(0.053)	(0.069)
Num.Obs.	150	150	150
R2 Adj.	0.431	0.848	0.849
R2 Within		0.202	0.201
R2 Within Adj.		0.142	0.140
RMSE	0.62	0.26	0.26
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

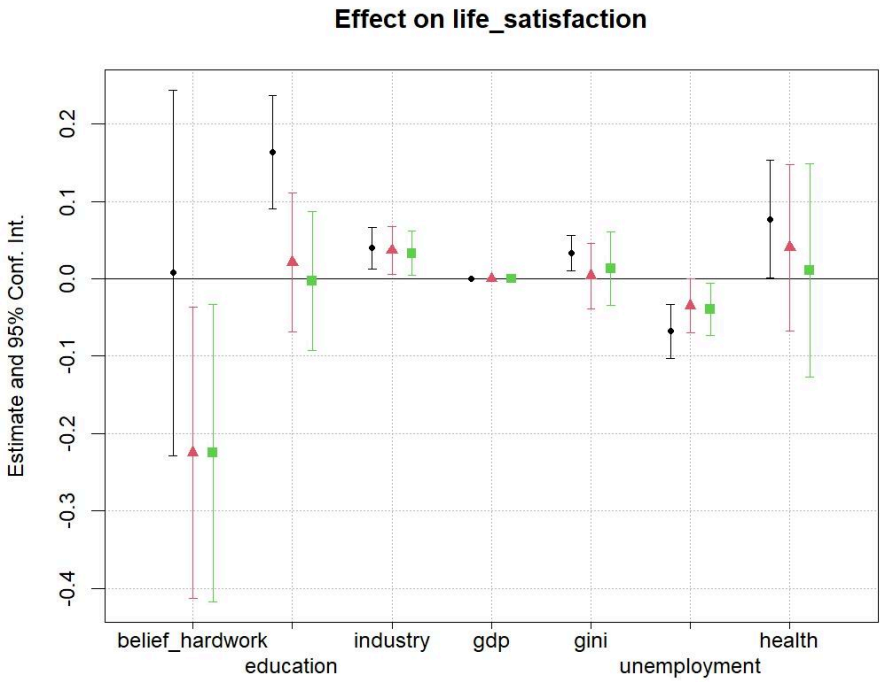
Results:

Pooled OLS: belief_hardwork = 0.007 (not significant, p > 0.1)

Country FE: belief_hardwork = -0.225* (p < 0.05)

Country+Wave FE: belief_hardwork = -0.225* (p < 0.05)

Figure 5: Confidence Intervals for Model 1.2



Interpretation:

Effect of Belief in Hard Work (belief_hardwork): The pattern holds. The effect is insignificant in pooled OLS but becomes significant ($p < 0.05$) and negative in both fixed effects models. The magnitude of the effect increases slightly compared to Models 1.0 and 1.1 (from -0.208 to -0.225), suggesting the relationship becomes more pronounced after controlling for economic structure.

Effect of New Control Variable (industry): The share of industrial employment has a consistent, positive, and statistically significant effect across all specifications (0.033* to 0.040**). This indicates that a larger industrial sector is associated with higher national life satisfaction, and this relationship holds even when analyzing changes within countries over time.

Effect of Education (education): As in Model 1.1, education shows a strong positive effect only in pooled OLS (0.164***) but becomes completely insignificant in fixed effects models, confirming its effect is driven by cross-country differences.

Effect of unemployment level: The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Test of Hypothesis for belief_hardwork:

H_0: $\beta_1 = 0$ — Rejected in both fixed effects models ($p < 0.05$).

H_1: $\beta_1 \neq 0$ — Supported. A statistically significant negative relationship persists between the collective belief in luck/connections and national life satisfaction, even after controlling for both education and the structure of the economy (industrial employment).

Conclusion: Adding the control for industrial employment (industry) does not eliminate the relationship between belief in hard work and life satisfaction. On the contrary, the link becomes slightly clearer. This means that a society's shared belief in the power of effort itself influences well-being, regardless of whether the country has a developed industrial sector or not.

Model 1_3 (Model 1_2 + personal_control)

Added variable: personal_control — the country-average level of perceived freedom of choice and control over one's life.

Model specification:

$$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot personal_control + \beta_3 \cdot education + \beta_4 \cdot industry + \beta_5 \cdot gdp + \beta_6 \cdot gini + \beta_7 \cdot unemployment + \beta_8 \cdot health + \varepsilon$$

Tests whether the previously significant effect of belief_hardwork disappears when controlling for the related psychological variable (personal_control).

Table 5: Estimated Regression Models for Life Satisfaction for Model 1.3

	Model 1 personal_control	**Model 1 with personal_control |FE1**	**Model 1 with with education, industry, personal_control|FE2**
(Intercept)	-1.622+		

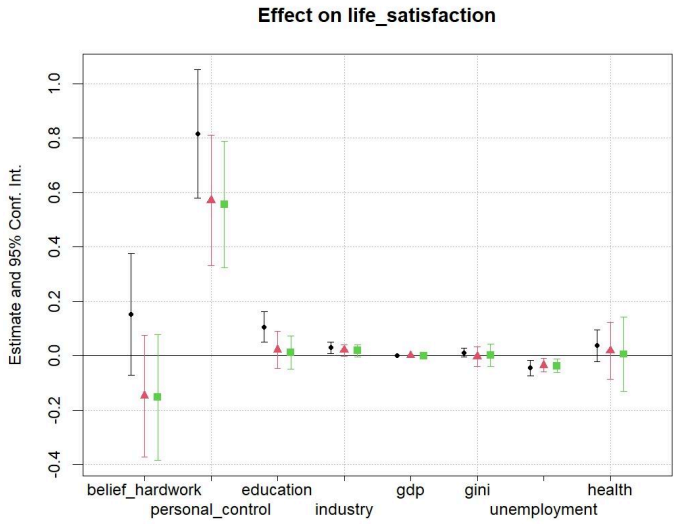
	(0.835)		
belief_hardwork	0.152	-0.148	-0.153
	(0.112)	(0.111)	(0.115)
personal_control	0.816***	0.570***	0.556***
	(0.118)	(0.119)	(0.116)
education	0.105***	0.022	0.011
	(0.028)	(0.034)	(0.030)
industry	0.029**	0.020+	0.018
	(0.010)	(0.011)	(0.011)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
gini	0.012	-0.004	0.001
	(0.008)	(0.018)	(0.021)
unemployment	-0.045**	-0.035**	-0.037**
	(0.014)	(0.012)	(0.012)
health	0.037	0.018	0.006
	(0.029)	(0.052)	(0.068)
Num.Obs.	150	150	150
R2 Adj.	0.670	0.889	0.888
R2 Within		0.426	0.413
R2 Within Adj.		0.376	0.361
RMSE	0.47	0.22	0.22
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

Pooled OLS: belief_hardwork = 0.152 (not significant, p > 0.1)

Country FE: belief_hardwork = -0.148 (not significant, p > 0.1)

Country+Wave FE: belief_hardwork = -0.153 (not significant, p > 0.1)

Figure 6: Confidence Intervals for Model 1.3



Interpretation:

The addition of personal_control fundamentally alters the finding. The previously significant negative effect of belief_hardwork observed in Models 1.0, 1.1, and 1.2 completely disappears and becomes statistically insignificant in all specifications of Model 1.3.

Effect of Personal Control (personal_control): This variable demonstrates an exceptionally strong, positive, and highly significant effect (0.556*** to 0.816***) across all models. It is the dominant predictor of national life satisfaction.

Effect of Other Controls: The effects of education and industry weaken considerably in the fixed effects models compared to previous specifications, becoming marginally significant or insignificant. The effect of unemployment remains negative and significant.

Effect of unemployment level: The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Test of Hypothesis for belief_hardwork:

H_0: $\beta_1 = 0$ — Cannot be rejected in any specification of Model 1.3.

H_1: $\beta_1 \neq 0$ — Not supported by the data in the final, most comprehensive model.

Conclusion: The perceived sense of personal control and freedom (personal_control) acts as a key confounding variable. It absorbs the explanatory power previously attributed to the belief in hard work. This indicates that the significant relationship found in earlier models was likely spurious: countries where people believe success comes from hard work also tend to be countries where people feel greater control over their lives. It is this latter factor — the feeling of agency — that primarily drives national differences in life satisfaction, not the specific cultural norm about the source of success. The hypothesis of an independent effect of belief in hard work on national well-being is not supported when this crucial psychological variable is accounted for.

For now and then we will check these hypotheses:

Hypothesis 1: Effect of Perceived Personal Control

H_0: β_personal_control = 0 (Perceived personal control does not affect life satisfaction)

H_1: β_personal_control ≠ 0 (Perceived personal control significantly affects life satisfaction)

Hypothesis 2: Effect of Belief in Hard Work

H_0: β_belief_hardwork = 0 (Belief in hard work does not affect life satisfaction after controlling for personal control)

H_1: β_belief_hardwork ≠ 0 (Belief in hard work retains a significant effect after controlling for personal control)

Testing Hypothesis 1 for Model 1.3: Effect of Perceived Personal Control

Pooled OLS: personal_control = 0.816*** ($p < 0.001$)

Country FE: personal_control = 0.570*** ($p < 0.001$)

Country+Wave FE: personal_control = 0.556*** ($p < 0.001$)

Statistical Significance: In all three model specifications, the coefficient for the variable personal_control is highly statistically significant at the $p < 0.001$ level . This means the probability of obtaining these results if the true effect were actually zero (if H_0 were true) is extremely low.

The coefficients are positive and have a large magnitude (from 0.556 to 0.816). This indicates a strong positive relationship: a one-point increase in the country-average level of perceived personal control is associated with an increase of 0.56 to 0.82 points in the average level of life satisfaction, all else being equal.

The significance and positive direction of the effect persist when moving from the simple model (Pooled OLS) to the more rigorous models with fixed effects, which control for time-invariant country characteristics and common time trends.

The null hypothesis H_0 (β_personal_control = 0) is confidently rejected.

The alternative hypothesis H_1 (β_personal_control ≠ 0) receives convincing statistical support.

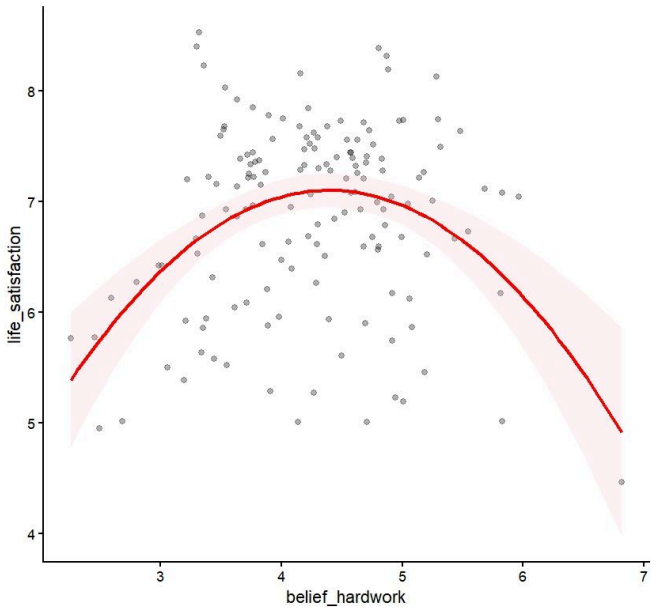
Interpretation: Perceived personal control is a statistically significant, strong, and robust predictor of the national level of life satisfaction. A higher collective sense of freedom of choice and control over one's life is associated with a higher level of well-being in a country.

MODEL 2

The purpose of the model is to test whether the relationship between belief in hard work and life satisfaction has a nonlinear (quadratic) form, not just a linear one.

Added the quadratic term I(belief_hardwork^2) to test for a U-shaped or inverted U-shaped dependency.

Figure 7: Confidence Intervals for Model 2



Regression equation:

life_satisfaction = $\beta_0 + \beta_1 \cdot$belief_hardwork + $\beta_2 \cdot$belief_hardwork^2 + $\beta_3 \cdot$personal_control + $\beta_4 \cdot$education + $\beta_5 \cdot$industry + $\beta_6 \cdot$gdp + $\beta_7 \cdot$gini + $\beta_8 \cdot$unemployment + $\beta_9 \cdot$health + ε

Analysis logic:

If $\beta_1 \neq 0$ and $\beta_2 = 0$ — a linear dependency is maintained

If $\beta_2 \neq 0$ and statistically significant — a quadratic dependency exists:

$\beta_2 > 0$ — U-shaped curve (satisfaction decreases, then increases)

$\beta_2 < 0$ — inverted U-shaped curve (satisfaction increases, then decreases)

Table 6: Estimated Regression Models for Life Satisfaction for Model 2

	Model 2	**Model 2|FE1**	**Model 2|FE2**
(Intercept)	-5.399***		
	(1.138)		
belief_hardwork	2.130***	0.818+	0.933*
	(0.519)	(0.419)	(0.395)
I(I(belief_hardwork^2))	-0.230***	-0.109*	-0.124**
	(0.061)	(0.044)	(0.040)

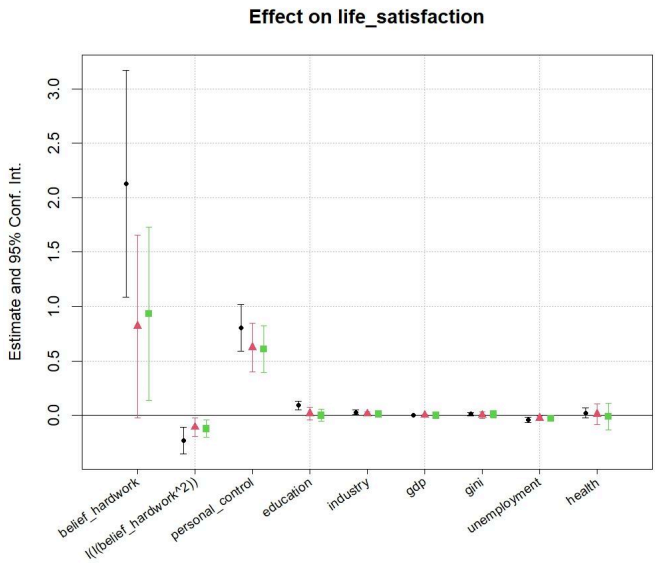
personal_control	0.804***	0.623***	0.608***
	(0.106)	(0.112)	(0.106)
education	0.091***	0.017	-0.001
	(0.021)	(0.029)	(0.027)
industry	0.028**	0.013	0.009
	(0.010)	(0.010)	(0.011)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
gini	0.011	0.001	0.008
	(0.007)	(0.015)	(0.017)
unemployment	-0.042***	-0.028*	-0.030*
	(0.012)	(0.012)	(0.011)
health	0.021	0.009	-0.012
	(0.023)	(0.047)	(0.061)
Num.Obs.	150	150	150
R2 Adj.	0.724	0.896	0.897
R2 Within		0.468	0.466
R2 Within Adj.		0.415	0.412
RMSE	0.43	0.21	0.21
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

Pooled OLS: belief_hardwork = 2.130***, I(belief_hardwork^2) = -0.230***

Country FE: belief_hardwork = 0.818+, I(belief_hardwork^2) = -0.109*

Country+Wave FE: belief_hardwork = 0.933*, I(belief_hardwork^2) = -0.124**

Figure 8: Confidence Intervals for Model 2



Interpretation:

1. Quadratic Effect Detected

The coefficient for the quadratic term I(belief_hardwork^2) is statistically significant and negative in all specifications. This indicates an inverted U-shaped relationship (∩-shaped curve) between belief in hard work and life satisfaction.

For an inverted parabola of the form $y = \beta_1 \cdot x + \beta_2 \cdot x^2$, the peak is calculated as: $x^* = -\beta_1/(2 \cdot \beta_2)$.

For the FE2 model: x* = -0.933/(2 × -0.124) = -0.933 / -0.248 = 3.76

This means that maximum life satisfaction is observed at a belief_hardwork value of approximately 3.76 on the 1-10 scale. Since the scale measures: 1 = "hard work" → 10 = "luck/connections", a value of 3.76 corresponds to a moderate belief in hard work.

Curve Interpretation:

At very strong belief in hard work (values 1-2) → life satisfaction is lower

At moderate belief in hard work (around 3.76) → life satisfaction is highest

At belief in luck/connections (values >5) → life satisfaction decreases

Effect of Fixed Effects:

In Pooled OLS: the effect is stronger (2.130*** and -0.230***)

With FE: the effect weakens but remains significant

The direction of the relationship (inverted U-shaped) persists

Effect of unemployment level: The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Conclusion: The relationship between collective beliefs about the sources of success and national well-being has a nonlinear nature. Moderate, rather than extreme, belief in the efficacy of hard work appears to be most favorable for life satisfaction.

MODEL 3 (Interaction with Education)

The purpose of the model is to test whether the effect of belief in hard work on life satisfaction changes depending on the country's level of education.

Added the interaction term belief_hardwork * education to test for moderation (conditional effect).

Regression equation:

$$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot education + \beta_3 \cdot (belief_hardwork \times education) + \beta_4 \cdot personal_control + \beta_5 \cdot industry + \beta_6 \cdot gdp + \beta_7 \cdot gini + \beta_8 \cdot unemployment + \beta_9 \cdot health + \varepsilon$$

Coefficient β_3 represents the interaction effect

If $\beta_3 \neq 0$ and statistically significant, then the influence of belief in hard work on satisfaction depends on the level of education

Interpretation: the effect of a one-unit change in belief_hardwork differs across different durations of education (education)

Table 7: Estimated Regression Models for Life Satisfaction for Model 3

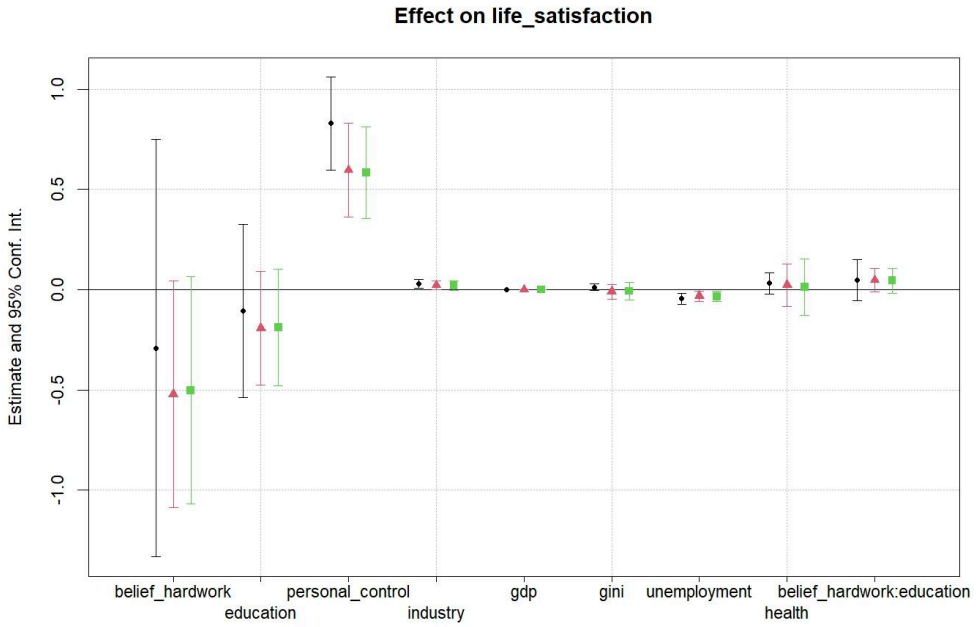
	Model 3	**Model 3|FE1**	**Model 3|FE2**
(Intercept)	0.247		
	(2.292)		
belief_hardwork	-0.291	-0.521+	-0.503+
	(0.519)	(0.281)	(0.282)
education	-0.107	-0.193	-0.188
	(0.214)	(0.141)	(0.145)
personal_control	0.830***	0.596***	0.584***
	(0.116)	(0.116)	(0.113)
industry	0.029**	0.022*	0.021+
	(0.011)	(0.011)	(0.012)
gdp	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
gini	0.012	-0.011	-0.007
	(0.008)	(0.018)	(0.021)
unemployment	-0.045**	-0.032*	-0.034**
	(0.014)	(0.013)	(0.012)
health	0.031	0.023	0.014
	(0.026)	(0.053)	(0.070)
belief_hardwork × education	0.048	0.047	0.045
	(0.051)	(0.030)	(0.031)
Num.Obs.	150	150	150
R2 Adj.	0.677	0.892	0.890
R2 Within		0.444	0.430
R2 Within Adj.		0.389	0.372
RMSE	0.46	0.22	0.22
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

Pooled OLS: belief_hardwork = -0.291 (not significant), education = -0.107 (not significant), belief_hardwork × education = 0.048 (not significant)

Country FE: belief_hardwork = -0.521+ ($p < 0.1$), education = -0.193 (not significant), belief_hardwork × education = 0.047 (not significant)

Country+Wave FE: belief_hardwork = -0.503+ ($p < 0.1$), education = -0.188 (not significant), belief_hardwork × education = 0.045 (not significant)

Figure 9: Confidence Intervals for Model 3



Interpretation:

Absence of Interaction Effect (Main Result):

The interaction coefficient belief_hardwork × education is not statistically significant in any of the model specifications ($p > 0.1$). This means that the effect of belief in hard work on life satisfaction does not depend on the country's level of education.

Unlike previous models where belief_hardwork was significant at $p < 0.05$, here its significance decreases to $p < 0.1$ in models with fixed effects. This is due to the addition of the interaction term, which redistributes explanatory power.

The education variable remains statistically insignificant in all specifications, consistent with previous findings that changes in education within countries are not associated with changes in life satisfaction.

The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Practical Interpretation:

The level of education does not strengthen or weaken the relationship between belief in hard work and well-being. The negative impact of a shift toward belief in luck/connections operates similarly in countries with both short and long education systems.

Conclusion: Education is not a moderator in the relationship between collective beliefs about sources of success and the national level of life satisfaction. The hypothesis that the cultural value of hard work matters more for well-being in more educated societies finds no support.

MODEL 4 (Interaction with Economic Development)

The purpose of the model is to test whether the effect of belief in hard work on life satisfaction changes depending on the country's level of economic development (GDP per capita).

Added the interaction term belief_hardwork * gdp to test for moderation (conditional effect).

Regression equation:

$$life_satisfaction = \beta_0 + \beta_1 \cdot belief_hardwork + \beta_2 \cdot gdp + \beta_3 \cdot (belief_hardwork \times gdp) + \beta_4 \cdot personal_control + \beta_5 \cdot education + \beta_6 \cdot industry + \beta_7 \cdot gini + \beta_8 \cdot unemployment + \beta_9 \cdot health + \varepsilon$$

Coefficient β_3 represents the interaction effect

If $\beta_3 \neq 0$ and statistically significant, then the influence of belief in hard work on satisfaction depends on the level of economic development

Practical question: Is the value of hard work more or less influential in rich versus poor countries?

Table 8: Estimated Regression Models for Life Satisfaction for Model 4

	Model 4	**Model 4|FE1**	**Model 4|FE2**
(Intercept)	-1.723*		
	(0.774)		
belief_hardwork	0.116	-0.225+	-0.222+
	(0.117)	(0.112)	(0.112)
gdp	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
personal_control	0.872***	0.544***	0.538***
	(0.110)	(0.116)	(0.116)
education	0.104***	0.022	0.016
	(0.027)	(0.036)	(0.032)
industry	0.026*	0.019+	0.019
	(0.010)	(0.010)	(0.012)
gini	0.010	-0.000	0.002
	(0.008)	(0.016)	(0.020)
unemployment	-0.039**	-0.034**	-0.035**
	(0.013)	(0.012)	(0.012)
health	0.020	0.017	0.011
	(0.029)	(0.052)	(0.067)
belief_hardwork × gdp	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
Num.Obs.	150	150	150
R2 Adj.	0.678	0.892	0.890
R2 Within		0.448	0.432

R2 Within Adj.		0.394	0.375
RMSE	0.46	0.22	0.22
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: COUNTRY		X	X
FE: wave			X

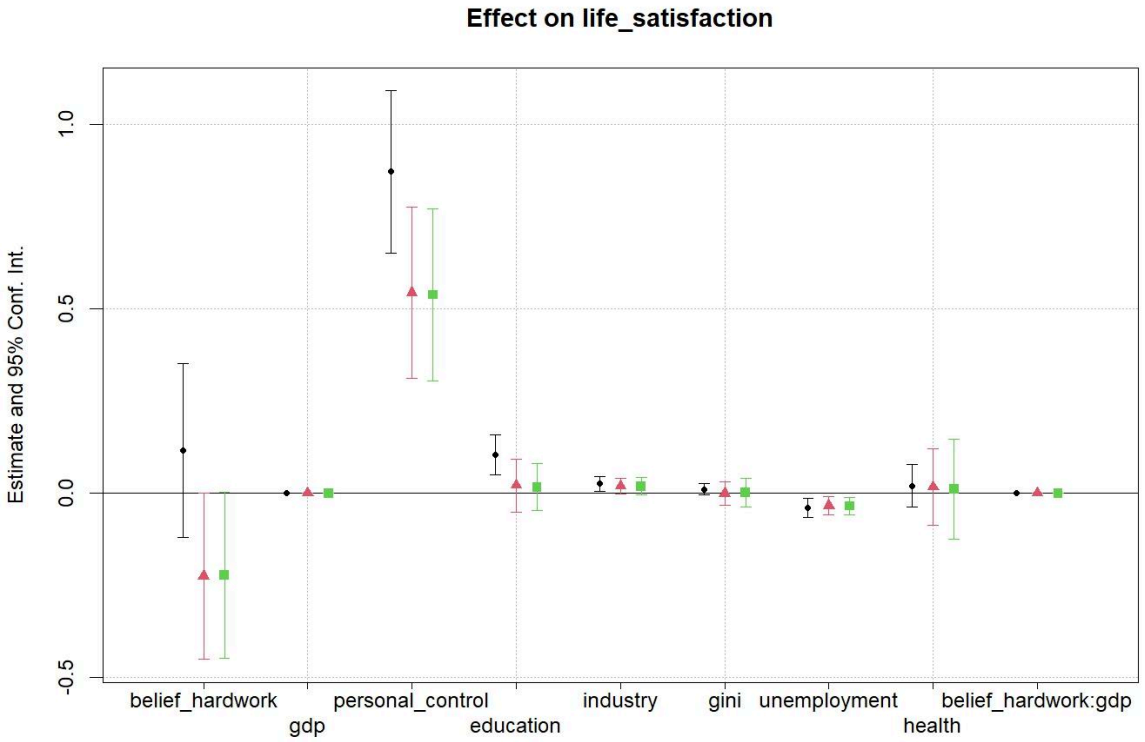
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Pooled OLS: belief_hardwork = 0.116 (not significant), gdp = -0.000 (not significant), belief_hardwork × gdp = 0.000 (not significant)

Country FE: belief_hardwork = -0.225+ (p < 0.1), gdp = -0.000 (not significant), belief_hardwork × gdp = 0.000 (not significant)

Country+Wave FE: belief_hardwork = -0.222+ (p < 0.1), gdp = -0.000 (not significant), belief_hardwork × gdp = 0.000 (not significant)

Figure 10: Confidence Intervals for Model 4



Interpretation:

1. Absence of Interaction Effect

The interaction coefficient belief_hardwork × gdp is not statistically significant in all model specifications (p > 0.1). This means that the effect of belief in hard work on life satisfaction does not depend on the country's level of economic development, measured by GDP per capita.

2. Main Effect of Belief in Hard Work:

As in previous models, belief_hardwork becomes statistically significant at the $p < 0.1$ level in models with fixed effects, confirming the within-country negative relationship between belief in luck/connections and life satisfaction.

3. Absence of Direct GDP Effect:

The gdp variable remains statistically insignificant in all specifications, consistent with previous findings that the absolute level of economic development is not a significant predictor of changes in life satisfaction within countries.

4. Practical Interpretation:

The effect of collective belief in hard work operates similarly in both rich and poor countries. The level of economic development does not strengthen or weaken the importance of this cultural value for national well-being. The negative impact of a shift toward belief in luck/connections manifests universally, regardless of GDP level.

5. The unemployment variable shows a stable, statistically significant negative effect on the national level of life satisfaction.

Conclusion: Economic development is not a moderator in the relationship between collective beliefs about sources of success and the national level of life satisfaction. The hypothesis that the cultural value of hard work holds different significance for well-being in richer versus poorer societies finds no empirical support.

All significant balanced models[3]

Table 9: Estimated Regression Results for Life Satisfaction in balanced models

	1.0 no FE	1.0 FE2	1.1 no FE	1.1 FE2	1.2 no FE	1.2 FE2	1.3 no FE	1.3 FE2	2 FE2
(Intercept)	5.327***		4.049***		3.144***		-1.622+		
	(0.828)		(0.738)		(0.722)		(0.835)		
belief_hardwork	0.049	-0.208*	0.051	-0.208*	0.007	-0.208*	0.152	-0.153	0.933*
	(0.146)	(0.099)	(0.123)	(0.100)	(0.118)	(0.100)	(0.112)	(0.115)	(0.395)
gdp	-0.000	0.000	-0.000	0.000	-0.000	0.000	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
gini	0.031*	0.007	0.025+	0.007	0.033**	0.007	0.012	0.001	0.008
	(0.014)	(0.022)	(0.013)	(0.025)	(0.011)	(0.025)	(0.008)	(0.021)	(0.017)
unemployment	-0.074***	-0.043**	-0.057**	-0.043**	-0.068***	-0.043**	-0.045**	-0.037**	-0.030*
	(0.020)	(0.016)	(0.020)	(0.016)	(0.017)	(0.016)	(0.014)	(0.012)	(0.011)
health	0.113**	-0.028	0.050	-0.028	0.077*	-0.028	0.037	0.006	-0.012
	(0.037)	(0.066)	(0.039)	(0.066)	(0.038)	(0.066)	(0.029)	(0.068)	(0.061)
education			0.177***	-0.001	0.164***	-0.001	0.105***	0.011	-0.001
			(0.049)	(0.047)	(0.036)	(0.047)	(0.028)	(0.030)	(0.027)

[3] The full table with all models is in the app because it's too big

industry					0.040**	0.033*	0.029**	0.018	0.009
					(0.013)	(0.014)	(0.010)	(0.011)	(0.011)
personal_control							0.816***	0.556***	0.608***
							(0.118)	(0.116)	(0.106)
I(I(belief_hardwork^2))									-0.124**
									(0.040)
Num.Obs.	150	150	150	150	150	150	150	150	150
R2 Adj.	0.225	0.847	0.360	0.846	0.431	0.846	0.670	0.888	0.897
R2 Within		0.176		0.176		0.176		0.413	0.466
R2 Within Adj.		0.131		0.122		0.122		0.361	0.412
RMSE	0.73	0.26	0.66	0.26	0.62	0.26	0.47	0.22	0.21
Std.Errors	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY	by: COUNTRY
FE: wave		X		X		X		X	X
FE: COUNTRY		X		X		X		X	X
· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001									

Comparing balanced and unbalanced data[4]

The same models are built for unbalanced data. In the table below, we compare the changes in the significance level of variables in models with unbalanced data compared to models with balanced data.

"Empty" - The significance did not change in unbalanced data compared to balanced data
"✖" - The significance was lost in the unbalanced data, but it was present in the balanced data.
"<" - The significance has increased
">" - The significance has decreased

(Intercept), gdp, belief_hardwork × education,belief_hardwork × gdp - didn't change the significance of any of the models.

Table 9.5: Comparing balanced and unbalanced data

	1.0 no FE	1.0 FE2	1.1 no FE	1.1 FE2	1.2 no FE	1.2 FE2	1.3 no FE	1.3 FE2	2 no FE	2 FE2	3 no FE	3 FE2	4 no FE	4 FE2
belief_hardwork		✖	✖	✖	✖	✖	>		<	✖			>	
gini	>			>										
unemployment				>			>	<	<		>			<
health						<								
education				<			<		<				<	
industry					<	✖	<		<		<			
personal_control	Very significant in all models in all data (***)													

[4] Full tables with all models and data are available in the application

I(I(belief_hardwork^2))									<	✖				

Main trends:

- The significance of personal control remains stable at 0.001 in all models
- The significance of hard work disappears or decreases in most models
- The significance of education and industry share remains significant in models without fixed effects, but their significance decreases
- The significance of a nonlinear relationship with a hardwork decreases and disappears
- The significance of the Gini coefficient increases in two models, and the significance of unemployment both decreases and increases

Conclusion:

In unbalanced data, there is a loss of significance for hard work and a decrease in significance for education and industry share, while personal control maintains a consistently high significance in all models. This is due to the bias in estimates when data is missing, but the variables that have little significance still have little significance, and the variables that were strongly significant are still strongly significant. Therefore, even if there is a bias in estimates in unbalanced data, the key significant factors are not affected and remain significant. We can conclude that in our data, life control has a consistently strong impact on life satisfaction.

General Conclusion of the Empirical Analysis

Analysis of panel data for 150 country-wave observations reveals several key patterns concerning the influence of the collective belief in the efficacy of hard work on the national level of life satisfaction.

1. Main Effect:

In the most rigorous specifications with country and time-period fixed effects, a statistically significant negative relationship ($p < 0.05$ in baseline Models 1.0-1.2) is found between belief in "luck/connections" (as opposed to hard work) and life satisfaction. This means that when a country, over time, increasingly believes that success depends on factors beyond personal control, its average level of well-being decreases.

Model 1.3 demonstrated that upon including the perceived personal control variable (personal_control), the effect of belief in hard work loses statistical significance at the standard 5% level, retaining only marginal significance ($p < 0.1$). This indicates that: Perceived personal control is the dominant predictor of national well-being (coefficient ~0.56***).

The effect of belief in hard work observed earlier is largely explained by a general psychological characteristic — the sense of control over one's life.

The cultural value "hard work leads to success" does not exert a strong independent influence on well-being after accounting for awareness of personal control.

2. Non-Linear Dependence

Model 2 revealed an inverted U-shaped relationship between belief in hard work and life satisfaction ($p < 0.05$ with FE). Maximum well-being is achieved with a moderate belief in hard work (value ~3.76 on a 1-10 scale), whereas both an extreme belief in hard work (values 1-2) and a belief in luck/connections (values >5) are associated with lower satisfaction.

3. Absence of Moderation by Structural Factors

Models 3 and 4 showed that the effect of belief in hard work:
-Does not depend on the country's level of education (interaction not significant).
-Does not depend on the level of economic development (interaction with GDP is not significant).
-This indicates the universality of the identified relationship, not contingent on socio-economic context.

4. Robust Macroeconomic Effects

Across all models, significant effects persist:
Unemployment exerts a consistently negative impact on well-being.
Education and GDP are significant only in cross-country comparisons, not in within-country dynamics.
Industrial employment has a weak positive effect.

Final Conclusion: People are happy not because they believe in hard work, but because they feel control over their lives. Unemployment consistently reduces well-being.

Economic Significance of the Findings

Prioritizing Human Agency: The results suggest that economic policy focused on expanding people's real opportunities and their perceived control over their lives (e.g., through quality education, protected labor rights, support for entrepreneurship) may have a more direct and stronger impact on collective well-being than measures aimed solely at increasing incomes or changing cultural attitudes toward work.

Unemployment as a Systemic Risk: The consistently significant negative effect of unemployment confirms its role as a key destabilizing factor for public well-being. This economically justifies the prioritization of employment and social protection policies, even during economic downturns, as they represent an investment in social stability.

Limitations of Macroeconomic Indicators: Even if a country's GDP grows, it does not guarantee that people will become happier. What matters is how the benefits of growth are distributed and whether citizens perceive tangible improvements in their own lives and opportunities.This calls into question the use of macroeconomic indicators (such as GDP) alone to assess the success of public policy.

An Optimal Value Balance: The identified non-linear relationship indicates that moderate, balanced belief systems are more conducive to societal well-being than extreme views that either completely deny or absolutize the role of personal effort in achieving success.

Critical analysis of the results obtained, analysis of the limitations of the study

There are a number of limitations in this study, which sets a vector for further expansion of the work. One of the limitations is the use of aggregated data. We analyzed the country-average indicators of life satisfaction, which allowed us to identify relationships at the macro level, but we did not take into account the country-wide heterogeneity. This does not allow us to see how the relationship between values and well-being differs between social groups. In addition, the sample size by country and wave is relatively small (150 observations), which limits statistical power in evaluating complex models (with interactions and nonlinearities) and increases the risk of coefficient instability.

The method of processing passes also imposes restrictions. In the balanced panel, individual passes were replaced by the average of the other waves, and countries with multiple passes were excluded. This can lead to a bias in estimates if omissions are not completely random (for example, they occur more often during periods of crisis or in countries with poor statistical quality). Comparing with an unbalanced panel partially reduces this problem, but it does not completely remove it.

Finally, the study remains observational: despite the use of fixed effects of countries and waves, we cannot unambiguously interpret the identified links as causal. Changes in the level of happiness can affect attitudes (happier people are more likely to believe in the "manageability" of life and the fairness of remuneration for work), and some important factors (the quality of institutions, political instability, cultural traditions) have not been measured or taken into account only indirectly.

Discussion part with suggestions for possible expansion of the study

Our results allow us to take a fresh look at the relationship between cultural attitudes and subjective well-being at the country level. This study was aimed at studying the belief in “success through work” as a potential resource for collective sustainability, however, the final models show that the key role is played not so much by faith in hard work as by a sense of personal control over life, people's conviction that their actions really affect the outcome of life. Where people believe that they can influence their own destiny, the level of life satisfaction is higher regardless of the level of economic development of the country.

Further development of the work can go in several directions. First, it is promising to move from purely aggregated models to a multi-level analysis combining individual WVS data and macro indicators of countries. This would allow us to check how the effects of faith in work and perceived control differ for different social groups within the same country (by gender, age, education, income, type of employment). Secondly, it makes sense to expand the set of cultural and institutional variables: to include indicators of trust, perception of justice, quality of justice, level of corruption, and political freedoms in order to separate the influence of values from the influence of institutions.

Third, we can explore other dimensions of well—being—emotional balance, trust in institutions, and satisfaction with certain areas of life-and check whether the same factors determine "overall happiness" and, for example, job or health satisfaction. Fourth, to strengthen the causal interpretation of the results, we should look for quasi-experimental situations: using external shocks (economic crises, labor market reforms, changes in social policy) or historical tools that influence attitudes and a sense of control, but are not directly related to current life satisfaction.

Assessing team members' contributions to group work

All team members contributed equally to the work.

Necessary references to sources

1. World Values Survey. (n.d.). *World Values Survey*. Retrieved October 24, 2024, from https://www.worldvaluessurvey.org/wvs.jsp
2. World Bank. (n.d.). *World Development Indicators (WDI)*. Retrieved November 15, 2024, from https://databank.worldbank.org/source/world-development-indicators

Links to required online applications

1. Used datasets: https://drive.google.com/drive/folders/1MJeoEkUWmWBsNNQMAo5NmXXjYtQNfQ1H?usp=drive_link
2. Link to the R code: https://drive.google.com/drive/folders/1TndHchbjwpemNrT9po4tFK1AWmyhgatd?usp=drive_link

Applications

Final table with all balanced models
since the table is too large, we can't insert it because it won't be readable, so we're attaching the file:
alltable_balanced.pdf

Final table with all unbalanced models
unbalanced_all.pdf

Application with codes and technical results
Regression Equation for Model 1

Code from R
For Balanced Data:

1. Excluding education, industry and personal_control

```
model1 <- feols(
  life_satisfaction ~ belief_hardwork + gdp + gini + unemployment +
health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY,
)
summary(model1)

model1_FE1 <- feols(
  life_satisfaction ~ belief_hardwork + gdp + gini + unemployment +
health | COUNTRY,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY,
)
summary(model1_FE1)

model1_FE2 <- feols(
  life_satisfaction ~ belief_hardwork + gdp + gini + unemployment +
health | COUNTRY + wave,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY,
)
summary(model1_FE2)
```

Technical results

Table 10: Technical results for model1

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	5.327477e+00	0.82770770	6.4364227762	4.9545e-08	***
belief_hardwork	4.934859e-02	0.14596849	0.3380769722	7.3675e-01	

gdp	-2.370000e-14	0.00000100	-0.0000000237	1.0000e+00	
gini	3.126518e-02	0.01415650	2.2085386464	3.1915e-02	*
unemployment	-7.384139e-02	0.02037847	-3.6235004908	6.8959e-04	***
health	1.130993e-01	0.03715358	3.0441010042	3.7475e-03	**

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.728602   Adj. R2: 0.225088
```

Table 11: Technical results for model1_FE1

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-1.977854e-01	0.09898731	-1.9980881502	0.051273	.
gdp	4.210000e-14	0.00000100	0.0000000421	1.000000	
gini	-8.356018e-03	0.01797468	-0.4648772033	0.644079	
unemployment	-3.805392e-02	0.01635800	-2.3263176893	0.024181	*
health	3.655699e-03	0.04847785	0.0754096880	0.940196	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.265166     Adj. R2: 0.844423
                 Within R2: 0.16687
```

Table 12: Technical results for model1_FE2

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-2.077208e-01	0.09939593	-2.0898319518	0.0418455	*
gdp	3.060000e-14	0.00000100	0.0000000306	1.0000000	
gini	7.322019e-03	0.02201109	0.3326512896	0.7408165	
unemployment	-4.329811e-02	0.01551635	-2.7904833490	0.0074787	**

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.259811     Adj. R2: 0.847432
                 Within R2: 0.175548
```

2. Including education

```
model1_1 <-feols(
  life_satisfaction ~ belief_hardwork + education + gdp + gini +
unemployment + health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_1)

model1_1_FE1 <-feols(
  life_satisfaction ~ belief_hardwork + education + gdp + gini +
unemployment + health |COUNTRY,
  data = FINAL_Merged_WVS_WBD_long,
```

```
  cluster = ~ COUNTRY
)
summary(model1_1_FE1)

model1_1_FE2 <-feols(
  life_satisfaction ~ belief_hardwork + education + gdp + gini +
unemployment + health | COUNTRY + wave,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_1_FE2)
```

Technical results

Table 13: Technical results for model1_1

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	4.048897e+00	0.73758475	5.489399e+00	1.4169e-06	***
belief_hardwork	5.086398e-02	0.12266650	4.146526e-01	6.8021e-01	
education	1.773471e-01	0.04887654	3.628471e+00	6.7923e-04	***
gdp	-6.550000e-15	0.00000100	-6.550000e-09	1.0000e+00	
gini	2.515607e-02	0.01262700	1.992244e+00	5.1931e-02	.
unemployment	-5.665137e-02	0.01975547	-2.867629e+00	6.0824e-03	**
health	4.982094e-02	0.03917100	1.271883e+00	2.0942e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.659824   Adj. R2: 0.360039
```

Table 14: Technical results for model1_1_FE1

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-2.018906e-01	0.09831757	-2.0534540175	0.045386	*
education	2.579764e-02	0.04703897	0.5484313060	0.585886	
gdp	4.340000e-14	0.00000100	0.0000000434	1.000000	
gini	-4.100803e-03	0.02263492	-0.1811715158	0.856980	
unemployment	-3.862753e-02	0.01640009	-2.3553238793	0.022554	*
health	9.330322e-05	0.05124279	0.0018208067	0.998555	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.264638     Adj. R2: 0.843394
                 Within R2: 0.170186
```

Table 15: Technical results for model1_1_FE2

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-2.075245e-01	0.10004891	-2.0742301945	0.0433333	*
education	-1.404268e-03	0.04700387	-0.0298755868	0.9762876	

gdp	3.050000e-14	0.00000100	0.0000000305	1.0000000	
gini	7.181093e-03	0.02502162	0.2869955699	0.7753259	
unemployment	-4.329839e-02	0.01558205	-2.7787349377	0.0077157	**
health	-2.783045e-02	0.06646604	-0.4187168216	0.6772528	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.259809     Adj. R2: 0.845775
                 Within R2: 0.175557
```

3. Including education and industry

```
model1_2 <- feols(
  life_satisfaction ~ belief_hardwork + education + industry + gdp +
gini + unemployment + health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_2)

model1_2_FE1 <- feols(
  life_satisfaction ~ belief_hardwork + education + industry + gdp +
gini + unemployment + health | COUNTRY,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_2_FE1)

model1_2_FE2 <-feols(
  life_satisfaction ~ belief_hardwork + education + industry + gdp +
gini + unemployment + health | COUNTRY + wave,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_2_FE2)
```

Technical results

Table 16: Technical results for model1_2

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	3.144460e+00	0.72173476	4.3568084366	6.7305e-05	***
belief_hardwork	7.343268e-03	0.11773581	0.0623707284	9.5052e-01	
education	1.638539e-01	0.03633659	4.5093357640	4.0644e-05	***
industry	3.951836e-02	0.01346965	2.9338821413	5.0806e-03	**
gdp	-3.030000e-14	0.00000100	-0.0000000303	1.0000e+00	
gini	3.294613e-02	0.01122677	2.9346045177	5.0706e-03	**
unemployment	-6.787037e-02	0.01731037	-3.9207934117	2.7434e-04	***

health	7.687603e-02	0.03800424	2.0228274970	4.8565e-02	*

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.619777   Adj. R2: 0.431388
```

Table 17: Technical results for model1_2_FE1

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-2.247480e-01	0.09370469	-2.3984709433	0.020316	*
education	2.084797e-02	0.04463635	0.4670625509	0.642525	
industry	3.643066e-02	0.01517711	2.4003684753	0.020222	*
gdp	3.070000e-14	0.00000100	0.0000000307	1.000000	
gini	3.560008e-03	0.02109114	0.1687916450	0.866655	
unemployment	-3.507051e-02	0.01716288	-2.0433931320	0.046410	*
health	4.023614e-02	0.05341921	0.7532148459	0.454925	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.259496     Adj. R2: 0.847801
                 Within R2: 0.202119
```

Table 18: Technical results for model1_2_FE2

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-2.251863e-01	0.09551559	-2.3575866828	0.022432	*
education	-2.891750e-03	0.04485008	-0.0644759250	0.948854	
industry	3.263487e-02	0.01417906	2.3016241782	0.025648	*
gdp	2.050000e-14	0.00000100	0.0000000205	1.000000	
gini	1.283430e-02	0.02370107	0.5415071512	0.590612	
unemployment	-3.969522e-02	0.01680134	-2.3626219372	0.022161	*
health	1.072869e-02	0.06880562	0.1559274921	0.876731	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.255695     Adj. R2: 0.848979
                 Within R2: 0.201459
```

4. Including education, industry and personal_control

```
model1_3 <- feols(life_satisfaction ~ belief_hardwork +
personal_control +  education + industry + gdp + gini + unemployment +
health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_3)

model1_3_FE1 <- feols(
  life_satisfaction ~ belief_hardwork + personal_control +  education +
industry + gdp + gini + unemployment + health | COUNTRY,
  data = FINAL_Merged_WVS_WBD_long,
```

```
  cluster = ~ COUNTRY
)
summary(model1_3_FE1)

model1_3_FE2 <- feols(
  life_satisfaction ~ belief_hardwork + personal_control +  education +
industry + gdp + gini + unemployment + health | COUNTRY + wave,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model1_3_FE2)
```

Technical results

Table 19: Technical results for model1_3

	Estimate	Std.	t value	Pr(>|t|)	
(Intercept)	-1.622498e+00	0.83482639	-1.943516e+00	5.7709e-02	.
belief_hardwork	1.519391e-01	0.11173675	1.359796e+00	1.8012e-01	
personal_control	8.157956e-01	0.11775211	6.928076e+00	8.5738e-09	***
education	1.050360e-01	0.02780493	3.777604e+00	4.2939e-04	***
industry	2.935093e-02	0.01011322	2.902233e+00	5.5383e-03	**
gdp	-7.730000e-15	0.00000100	-7.730000e-09	1.0000e+00	
gini	1.154615e-02	0.00823033	1.402877e+00	1.6696e-01	
unemployment	-4.541081e-02	0.01429200	-3.177358e+00	2.5733e-03	**
health	3.650660e-02	0.02863399	1.274940e+00	2.0834e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.470145   Adj. R2: 0.670483
```

Table 20: Technical results for model1_3_FE1

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-1.477183e-01	0.11138590	-1.3261849287	1.9093e-01	
personal_control	5.700562e-01	0.11909251	4.7866672671	1.6013e-05	***
education	2.206459e-02	0.03370832	0.6545740731	5.1580e-01	
industry	1.982150e-02	0.01077276	1.8399638031	7.1835e-02	.
gdp	4.170000e-14	0.00000100	0.0000000417	1.0000e+00	
gini	-3.561634e-03	0.01766013	-0.2016766162	8.4100e-01	
unemployment	-3.523062e-02	0.01224857	-2.8763048493	5.9415e-03	**
health	1.798598e-02	0.05177339	0.3473982142	7.2978e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.220173     Adj. R2: 0.889242
                 Within R2: 0.42561
```

Table 21: Technical results for model1_3_FE2

	Estimate	Std.	t value	Pr(>|t|)	
belief_hardwork	-1.530195e-01	0.11468980	-1.3342027816	1.8831e-01	
personal_control	5.555192e-01	0.11581365	4.7966645526	1.5479e-05	***
education	1.126969e-02	0.03034367	0.3714017738	7.1194e-01	
industry	1.829070e-02	0.01137482	1.6079994472	1.1426e-01	
gdp	3.680000e-14	0.00000100	0.0000000368	1.0000e+00	
gini	7.703460e-04	0.02053872	0.0375070032	9.7023e-01	
unemployment	-3.721901e-02	0.01199857	-3.1019543512	3.1865e-03	**
health	5.518080e-03	0.06790078	0.0812668072	9.3556e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.219215     Adj. R2: 0.887764
                 Within R2: 0.413062
```

Similarly For Unbalanced Data

Regression Equation for Model 2

Code from R

```
model2 <- feols(life_satisfaction ~ belief_hardwork +
I(belief_hardwork^2) + personal_control +  education + industry+ gdp +
gini + unemployment + health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model2)


model2_FE1 <- feols(life_satisfaction ~ belief_hardwork +
I(belief_hardwork^2) + personal_control +  education + industry+ gdp +
gini + unemployment + health | COUNTRY,
                data = FINAL_Merged_WVS_WBD_long,
                cluster = ~ COUNTRY
)
summary(model2_FE1)

model2_FE2 <- feols(life_satisfaction ~ belief_hardwork +
I(belief_hardwork^2) + personal_control +  education + industry+ gdp +
gini + unemployment + health | COUNTRY + wave,
                    data = FINAL_Merged_WVS_WBD_long,
                    cluster = ~ COUNTRY
)
summary(model2_FE2)
```

Technical results

Table 22: Technical results for model2

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	-5.398713e+00	1.13755128	-4.745907e+00	1.8382e-05	***
belief_hardwork	2.129959e+00	0.51916603	4.102654e+00	1.5376e-04	***
I(I(belief_hardwork^2))	-2.297447e-01	0.06107199	-3.761868e+00	4.5085e-04	***
personal_control	8.043605e-01	0.10593945	7.592644e+00	8.0571e-10	***
education	9.097804e-02	0.02058098	4.420491e+00	5.4564e-05	***
industry	2.800222e-02	0.00991569	2.824033e+00	6.8385e-03	**
gdp	-6.180000e-15	0.00000100	-6.180000e-09	1.0000e+00	
gini	1.099564e-02	0.00685435	1.604184e+00	1.1510e-01	
unemployment	-4.171143e-02	0.01187490	-3.512570e+00	9.6430e-04	***
health	2.132017e-02	0.02340293	9.110041e-01	3.6675e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.428553   Adj. R2: 0.724251
```

Table 23: Technical results for model2_FE1

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	8.179194e-01	0.41858186	1.9540250076	5.6418e-02	.
I(I(belief_hardwork^2))	-1.091734e-01	0.04383997	-2.4902713958	1.6204e-02	*
personal_control	6.233628e-01	0.11225869	5.5529135981	1.1341e-06	***
education	1.693871e-02	0.02939330	0.5762778884	5.6707e-01	
industry	1.317359e-02	0.00981618	1.3420278198	1.8577e-01	
gdp	4.570000e-14	0.00000100	0.0000000457	1.0000e+00	
gini	9.608860e-04	0.01462531	0.0657002026	9.4788e-01	
unemployment	-2.765819e-02	0.01155108	-2.3944250501	2.0517e-02	*
health	8.945745e-03	0.04742376	0.1886342480	8.5116e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.211957     Adj. R2: 0.896226
                 Within R2: 0.467679
```

Table 24: Technical results for model2_FE2

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	9.327642e-01	0.39529958	2.3596387867	2.2321e-02	*
I(I(belief_hardwork^2))	-1.237022e-01	0.03952332	-3.1298540299	2.9452e-03	**
personal_control	6.078907e-01	0.10610753	5.7290064520	6.1039e-07	***
education	-8.912273e-04	0.02683835	-0.0332072333	9.7364e-01	

industry	9.446765e-03	0.01083065	0.8722248448	3.8734e-01	
gdp	3.850000e-14	0.00000100	0.0000000385	1.0000e+00	
gini	8.345384e-03	0.01663360	0.5017185670	6.1811e-01	
unemployment	-2.963261e-02	0.01138881	-2.6019064685	1.2227e-02	*
health	-1.150271e-02	0.06089689	-0.1888883693	8.5096e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.209148     Adj. R2: 0.896688
                 Within R2: 0.465733
```

Regression Equation for Model 3

Code from R

```
model3 <- feols(
  life_satisfaction ~ belief_hardwork * education + personal_control +
industry + gdp + gini + unemployment + health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)

summary(model3)

model3_FE1 <- feols(
  life_satisfaction ~ belief_hardwork * education + personal_control +
industry + gdp + gini + unemployment + health | COUNTRY,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model3_FE1)

model3_FE2 <- feols(
  life_satisfaction ~ belief_hardwork * education + personal_control +
industry + gdp + gini + unemployment + health | COUNTRY + wave,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)

summary(model3_FE2)
```

Technical results

Table 25: Technical results for model3

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	2.465316e-01	2.29160469	1.075803e-01	9.1477e-01	
belief_hardwork	-2.913496e-01	0.51912933	-5.612274e-01	5.7720e-01	

education	-1.065959e-01	0.21449290	-4.969671e-01	6.2143e-01	
personal_control	8.303968e-01	0.11556298	7.185664e+00	3.4227e-09	***
industry	2.852058e-02	0.01054300	2.705168e+00	9.3642e-03	**
gdp	-3.040000e-15	0.00000100	-3.040000e-09	1.0000e+00	
gini	1.170042e-02	0.00809635	1.445146e+00	1.5478e-01	
unemployment	-4.473082e-02	0.01365335	-3.276179e+00	1.9367e-03	**
health	3.148845e-02	0.02624739	1.199679e+00	2.3603e-01	
belief_hardwork: education	4.834949e-02	0.05082970	9.512055e-01	3.4617e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.464156   Adj. R2: 0.67653
```

Table 26: Technical results for model3_FE1

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-5.214350e-01	0.28129698	-1.8536814785	6.9810e-02	.
education	-1.929358e-01	0.14068122	-1.3714396594	1.7649e-01	
personal_control	5.961095e-01	0.11609207	5.1347998392	4.8645e-06	***
industry	2.201446e-02	0.01064512	2.0680329071	4.3937e-02	*
gdp	3.910000e-14	0.00000100	0.0000000391	1.0000e+00	
gini	-1.066334e-02	0.01767483	-0.6033064033	5.4909e-01	
unemployment	-3.184323e-02	0.01256318	-2.5346471252	1.4501e-02	*
health	2.308088e-02	0.05271126	0.4378737387	6.6340e-01	
belief_hardwork: educati	on 4.740863e-02	0.02999889	1.5803461450	1.2046e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.216588     Adj. R2: 0.891642
                 Within R2: 0.444163
```

Table 27: Technical results for model3_FE2

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-5.027883e-01	0.28210952	-1.7822450165	8.0906e-02	.
education	-1.883810e-01	0.14495383	-1.2995932253	1.9982e-01	
personal_control	5.839323e-01	0.11318745	5.1589849598	4.4746e-06	***
industry	2.084674e-02	0.01182809	1.7624762011	8.4226e-02	.
gdp	3.570000e-14	0.00000100	0.0000000357	1.0000e+00	
gini	-7.083104e-03	0.02125727	-0.3332084976	7.4040e-01	
unemployment	-3.351724e-02	0.01217997	-2.7518319157	8.2847e-03	**
health	1.361334e-02	0.06980319	0.1950246164	8.4618e-01	
belief_hardwork: educati	on 4.467192e-02	0.03095976	1.4429026109	1.5541e-01	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.216106     Adj. R2: 0.889699
                 Within R2: 0.429592
```

Model 4: Interaction with Economic Development

Code from R

```
model4 <- feols(life_satisfaction ~ belief_hardwork * gdp +
personal_control +  education + industry + gini + unemployment +
health,
  data = FINAL_Merged_WVS_WBD_long,
  cluster = ~ COUNTRY
)
summary(model4)

model4_FE1 <- feols(life_satisfaction ~ belief_hardwork * gdp +
personal_control +  education + industry + gini + unemployment + health
| COUNTRY,
                data = FINAL_Merged_WVS_WBD_long,
                cluster = ~ COUNTRY
)
summary(model4_FE1)

model4_FE2 <- feols(life_satisfaction ~ belief_hardwork * gdp +
personal_control +  education + industry + gini + unemployment + health
| COUNTRY + wave,
                    data = FINAL_Merged_WVS_WBD_long,
                    cluster = ~ COUNTRY
)
summary(model4_FE2)
```

Technical results

Table 28: Technical results for model4

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	-1.722800e+00	0.77419682	-2.2252737341	3.0697e-02	*
belief_hardwork	1.158800e-01	0.11731864	0.9877376705	3.2813e-01	
gdp	-3.560000e-13	0.00000100	-0.0000003560	1.0000e+00	
personal_control	8.716055e-01	0.10985455	7.9341774723	2.4090e-10	***
education	1.040129e-01	0.02722994	3.8197978069	3.7657e-04	***
industry	2.578348e-02	0.01010341	2.5519591016	1.3881e-02	*
gini	1.043688e-02	0.00778508	1.3406258567	1.8622e-01	
unemployment	-3.903405e-02	0.01305711	-2.9894860179	4.3608e-03	**
health	1.996961e-02	0.02854797	0.6995108221	4.8754e-01	
belief_hardwork: gdp	9.250000e-14	0.00000100	0.0000000925	1.0000e+00	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.463125   Adj. R2: 0.677966
```

Table 29: Technical results for model4_FE1

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-2.249828e-01	0.11223694	-2.004534578	5.0555e-02	.
gdp	-4.840000e-13	0.00000100	-0.000000484	1.0000e+00	
personal_control	5.437758e-01	0.11580805	4.695492154	2.1792e-05	***
education	2.155597e-02	0.03590936	0.600288539	5.5108e-01	
industry	1.947933e-02	0.01048556	1.857728239	6.9222e-02	.
gini	-3.220576e-04	0.01619651	-0.019884380	9.8422e-01	
unemployment	-3.367159e-02	0.01198950	-2.808422802	7.1298e-03	**
health	1.730100e-02	0.05184914	0.333679694	7.4005e-01	
belief_hardwork: gdp	1.400000e-13	0.00000100	0.000000140	1.0000e+00	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.215803     Adj. R2: 0.892426
                  Within R2: 0.448185
```

Table 30: Technical results for model4_FE2

	Estimate	Std. Error	t value	Pr(>|t|)	
belief_hardwork	-2.219559e-01	0.11218096	-1.978552551	5.3502e-02	.
gdp	-4.560000e-13	0.00000100	-0.000000456	1.0000e+00	
personal_control	5.380560e-01	0.11557482	4.655477317	2.4933e-05	***
education	1.631839e-02	0.03151442	0.517807185	6.0692e-01	
industry	1.884740e-02	0.01157383	1.628449937	1.0984e-01	
gini	1.630099e-03	0.01960572	0.083144039	9.3408e-01	
unemployment	-3.477136e-02	0.01160123	-2.997212184	4.2687e-03	**
health	1.112703e-02	0.06723626	0.165491482	8.6924e-01	
belief_hardwork: gdp	1.330000e-13	0.00000100	0.000000133	1.0000e+00	

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.215601     Adj. R2: 0.890215
                  Within R2: 0.432257
```

Creating summary models for data presentation

Code from R

```
modelsummary(list("Model 1" = model1, "Model 1|Country" = model1_FE1,
"Model 1|Country + wawe" = model1_FE2 ),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
```

```
             output = "table1FE2.docx")
modelsummary(list("Model 1 with education" = model1_1,
                  "Model 1 with education|FE1" = model1_1_FE1,
                  "Model 1 with education|FE2" = model1_1_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table1_1_FE2.docx")
modelsummary(list("Model 1 with education and industry" = model1_2,
                  "Model 1 with education and industry|FE1" =
model1_2_FE1,
                  "Model 1 with education and industry|FE2" =
model1_2_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table1_2_FE2.docx")
modelsummary(list(
                  "Model 1 with education and industry" = model1_3,
                  "Model 1 with education and industry|FE1" =
model1_3_FE1,
                  "Model 1 with with education, industry,
personal_control|FE2" = model1_3_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table1_3_FE2.docx")
modelsummary(list("Model 2" = model2,
                  "Model 2|FE1" = model2_FE1,
                  "Model 2|FE2" = model2_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table2_FE2.docx")
modelsummary(list("Model 3" = model3,
                  "Model 3|FE1" = model3_FE1,
                  "Model 3|FE2" = model3_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table3_FE2.docx")
modelsummary(list("Model 4" = model4,
                  "Model 4|FE1" = model4_FE1,
                  "Model 4|FE2" = model4_FE2),
             stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "table4_FE2.docx")
panels <- list(
  "Model 1" = list(
```

```
    "1.0 no FE" = model1,
    "1.0 FE1" = model1_FE1,
    "1.0 FE2" = model1_FE2),
  "Model 1_1" = list(
    "1.1 no FE" = model1_1,
    "1.1 FE1" = model1_1_FE1,
    "1.1 FE2" = model1_1_FE2),
  "Model 1_2" = list(
    "1.2 no FE" = model1_2,
    "1.2 FE1" = model1_2_FE1,
    "1.2 FE2" = model1_2_FE2),
  "Model 1_3" = list(
    "1.3 no FE" = model1_3,
    "1.3 FE1" = model1_3_FE1,
    "1.3 FE2" = model1_3_FE2),
  "Model 2" = list(
    "2 no FE" = model2,
    "2 FE1" = model2_FE1,
    "2 FE2" = model2_FE2),
  "Model 3" = list(
    "3 no FE" = model3,
    "3 FE1" = model3_FE1,
    "3 FE2" = model3_FE2),

  "Model 4" = list(
    "4 no FE" = model4,
    "4 FE1" = model4_FE1,
    "4 FE2" = model4_FE2))

    # shape = "cbind"
modelsummary(panels, shape = "cbind", stars = TRUE,
             gof_omit = "R2$|AIC|BIC"
            )

panels1 <- list(
  "Model 1" = list(
    "1.0 no FE" = model1,
    "1.0 FE2" = model1_FE2),
  "Model 1_1" = list(
    "1.1 no FE" = model1_1,
    "1.1 FE2" = model1_1_FE2),
  "Model 1_2" = list(
    "1.2 no FE" = model1_2,
```

```
    "1.2 FE2" = model1_2_FE2),
  "Model 1_3" = list(
    "1.3 no FE" = model1_3,

    "1.3 FE2" = model1_3_FE2),
  "Model 2" = list(
   "2 FE2" = model2_FE2))


# shape = "cbind"
modelsummary(panels1, shape = "cbind", stars = TRUE,
             gof_omit = "R2$|AIC|BIC",
             output = "importanttable1.docx")
#unbalanced data
panels2 <- list(
  "Model 1" = list(
    "1.0 no FE" = model1,
    "1.0 FE1" = model1_FE1,
    "1.0 FE2" = model1_FE2),
  "Model 1_1" = list(
    "1.1 no FE" = model1_1,
    "1.1 FE1" = model1_1_FE1,
    "1.1 FE2" = model1_1_FE2),
  "Model 1_2" = list(
    "1.2 no FE" = model1_2,
    "1.2 FE1" = model1_2_FE1,
    "1.2 FE2" = model1_2_FE2),
  "Model 1_3" = list(
    "1.3 no FE" = model1_3,
    "1.3 FE1" = model1_3_FE1,
    "1.3 FE2" = model1_3_FE2),
  "Model 2" = list(
    "2 no FE" = model2,
    "2 FE1" = model2_FE1,
    "2 FE2" = model2_FE2),
  "Model 3" = list(
    "3 no FE" = model3,
    "3 FE1" = model3_FE1,
    "3 FE2" = model3_FE2),

  "Model 4" = list(
    "4 no FE" = model4,
    "4 FE1" = model4_FE1,
    "4 FE2" = model4_FE2))
```

```
# shape = "cbind"
modelsummary(panels2, shape = "cbind", stars = TRUE,
             gof_omit = "R2$|AIC|BIC"
)
```